

USDe: A Crypto-Native Synthetic Dollar

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Abstract

USDe is a crypto-native synthetic dollar issued by the Ethena protocol. It maintains a 1:1 peg to the US dollar not through fiat reserves held in a bank, but through a delta-neutral hedging strategy: spot crypto collateral balanced against equivalent short perpetual futures positions. The combination eliminates directional price exposure while generating real yield from three independent sources: perpetual funding rates, Ethereum staking rewards, and returns on liquid stablecoin reserves. This paper describes USDe's design architecture, minting and redemption mechanics, yield distribution system, off-chain hedging infrastructure, risk surface, and the reserve fund that provides a backstop against adverse funding conditions.

Metric	Value (as of Q4 2025)
USDe circulating supply	~\$10.5 billion
Market cap rank (stablecoins)	3rd largest globally
sUSDe yield (historical avg 2024)	~19% APY
Chains supported	24+
Reserve fund size	>\$41 million
Collateral custody model	Off-exchange settlement (OES)

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The Problem with Existing Stablecoins

Every stablecoin design makes a tradeoff. The three dominant models each carry a structural cost that limits how far they can scale or how trustless they can actually be.

Fiat-backed stablecoins (USDC, USDT)

Reserves held in banks and short-term treasuries give these stablecoins their stability, but that same structure creates a direct dependency on the traditional financial system. Accounts can be frozen. Issuers can be regulated out of existence in a jurisdiction. The dollar holdings are custodied by institutions that operate under government oversight, making fiat-backed stablecoins censorship-resistant in name only. They also generate yield entirely off-chain, which the issuer typically retains rather than distributing to holders.

Over-collateralized crypto-backed stablecoins (DAI, crvUSD)

These designs achieve greater decentralization by using on-chain crypto assets as collateral. The trade-off is capital inefficiency. A user must lock \$150 of ETH to mint \$100 of stablecoin. This overcollateralization requirement caps the total addressable supply, because the system cannot grow beyond the willingness of users to tie up excess capital. These stablecoins also inherit the volatility of their collateral in the form

of liquidation risk, and their yields are modest relative to the capital tied up.

Algorithmic stablecoins (UST/LUNA)

Algorithmic designs attempt to use reflexive tokenomics instead of real collateral to maintain a peg. The Terra/LUNA collapse in May 2022 demonstrated what happens when the feedback loop driving these designs runs in reverse. An algorithmic stablecoin that relies on demand for its governance token to maintain the peg of its dollar token is not backed by anything other than market confidence. When that confidence breaks, the death spiral is self-reinforcing. The category has not recovered from this event.

The design space has been looking for a stablecoin that is simultaneously: fully collateralized, crypto-native (not dependent on bank accounts), scalable without requiring overcollateralization, and yield-generating for holders. USDe is Ethena's answer to that problem.

What USDe Is

USDe is a synthetic dollar: a token whose USD value is maintained not by holding dollars, but by constructing a position whose net dollar value is always approximately one dollar, regardless of what happens to the underlying crypto assets.

The construction works as follows. When a user deposits \$100 of ETH or stETH into the Ethena protocol, two things happen simultaneously. The protocol holds the deposited ETH in off-chain custody (described in Section 4). It also opens a short perpetual futures position on a centralized derivatives exchange for the same notional dollar value, \$100 of ETH exposure. The short position profits when ETH price falls. The spot collateral loses value when ETH price falls. The gains and losses cancel out, leaving the combined position worth approximately \$100 regardless of ETH's price movements. This is delta neutrality.

Because the position is perfectly hedged, USDe only requires 1:1 collateralization. There is no overcollateralization buffer needed, because there is no directional price risk

to buffer against. This is the source of USDe's capital efficiency advantage over crypto-backed stablecoins.

Property	Fiat-backed	Over-collateralized	USDe (Synthetic)
Collateral ratio	1:1 (fiat)	150%+ (crypto)	1:1 (hedged)
Censorship resistance	Low	High	Medium-High
Yield to holder	None (issuer keeps)	Low-moderate	High (funding+staking)
Banking dependency	Yes	No	No
Scalability	High	Limited	High

Table 1: Stablecoin model comparison

Delta-Neutral Mechanics

Delta, in options and derivatives terminology, measures how much the value of a position changes for a given change in the underlying asset's price. A delta of +1 means the position gains \$1 for every \$1 increase in the underlying. A delta of -1 means it loses \$1. A delta-neutral position has a net delta of approximately zero: price movements in the underlying have minimal effect on the total position value.

Constructing the hedge

Suppose ETH is trading at \$3,000. A user deposits 1 ETH (worth \$3,000) into Ethena. The protocol immediately opens a short perpetual futures position for 1 ETH of notional value on a derivatives exchange.

If ETH rises to \$4,000: the spot collateral is now worth \$4,000 (+\$1,000), but the short position has lost \$1,000. Net change: \$0. The position is still worth \$3,000 in USD terms.

If ETH falls to \$2,000: the spot collateral is now worth \$2,000 (-\$1,000), but the short position has gained \$1,000. Net change: \$0. The position is still worth \$3,000 in USD terms.

In both cases, the combined position holds its dollar value. This is what allows Ethena to issue \$3,000 of USDe against \$3,000 of collateral without any buffer: the hedge removes the price risk that would otherwise require over-collateralization.

Why perpetual futures specifically

Perpetual futures (perps) are the dominant derivatives instrument in crypto. Unlike dated futures contracts, perps have no expiry date. Instead, they use a funding rate mechanism to keep the contract price anchored to the spot price. When the perp trades above the spot price, longs pay shorts. When it trades below, shorts pay longs. Historically, funding rates in crypto markets have been persistently positive, meaning longs pay shorts. Because Ethena is always short, the protocol collects funding payments as long as this condition holds. This is one of USDe's primary yield sources.

Basis trading as a secondary hedge instrument

In addition to perpetual futures, Ethena uses dated futures contracts (calendar futures) for basis trading. When a dated futures contract trades at a premium to spot (a condition called contango), the protocol can capture that premium by holding spot and selling the futures contract. As the contract approaches expiry and converges to spot, the spread is earned as profit. This is a well-understood strategy in traditional commodity and fixed income markets, adapted here for crypto.

Collateral Composition and Off-Exchange Custody

USDe is backed by four categories of assets, with the mix shifting dynamically based on market conditions and available yield opportunities:

- Staked ETH (stETH, via Lido): earns Ethereum consensus and execution layer rewards in addition to serving as hedge collateral.
- ETH (unstaked): used as collateral on exchanges that do not accept staked

ETH as margin. • BTC and cbBTC: broadens the collateral base and hedge capacity.

- Liquid stablecoins (USDC, USDT, USDtb): deployed during bear markets or when funding rates are negative, earning money-market or T-bill yields instead.

Off-Exchange Settlement (OES)

A natural concern with any strategy that involves centralized exchanges is counterparty risk. If Ethena held collateral on a derivatives exchange and that exchange became insolvent (as FTX did in November 2022), the collateral would be lost.

Ethena avoids this by never actually depositing collateral onto exchanges. Instead, backing assets are held in institutional-grade Off-Exchange Settlement custody solutions. Ethena delegates signing authority to exchanges (allowing them to margin the short positions) but never transfers beneficial ownership of the assets. Settlement of P&L; and funding payments flows between custody and the exchange in cash, not by moving collateral. The collateral itself remains in custody, outside exchange reach, at all times.

This structure meant that when the Bybit exchange was hacked in February 2025, Ethena's collateral was unaffected. The protocol's OES providers held the assets; the exchange exposure was limited to unmarked P&L; in transit.

Minting and Redemption

USDe has two access tiers, designed to balance permissionless composability with the operational requirements of maintaining the hedge.

Whitelisted minting and redemption

Approved parties (institutional market makers who have passed KYC/KYB screening) can mint and redeem USDe directly with the Ethena contracts. The process is atomic: a whitelisted user provides approximately \$100 of eligible collateral and receives approximately 100 USDe in a single transaction, minus execution and gas costs. Ethena

earns no profit on minting or redemption; slippage and execution fees are included in the price returned.

On minting, the protocol opens a corresponding short perpetual position for the same notional value as the deposited collateral. On redemption, it closes that position and returns the collateral. The hedge is always maintained in proportion to outstanding USDe supply.

Permissionless secondary market access

Non-whitelisted users can acquire USDe by swapping through external AMM liquidity pools (primarily on Curve and Uniswap) where USDe/USDC and USDe/USDT pairs trade. The arbitrage mechanism enforced by whitelisted parties keeps secondary market prices aligned with the \$1 peg: if USDe trades below \$1 on a DEX, a whitelisted party can buy it cheaply and redeem at par for profit. If it trades above \$1, they can mint at \$1 and sell for profit. This two-sided arbitrage pressure continuously corrects deviations.

Peg stability in practice

During the October 2025 flash crash, USDe showed a brief depeg on certain secondary market venues, reportedly as low as \$0.65 on Binance's trading pair in isolated instances. However, on Curve (the primary liquidity venue for USDe), the price remained close to \$1, and Ethena processed large redemptions at par. Post-incident analysis attributed the Binance deviation to an oracle lag and low on-venue liquidity rather than a breakdown of the underlying backing. The protocol processed all redemption requests without pausing, and the peg fully recovered.

Yield Generation and sUSDe

USDe is not a passive instrument. The protocol generates revenue from three independent sources, and that revenue is distributed to sUSDe holders.

Source 1: Perpetual funding rates

The primary yield driver. In crypto derivatives markets, there is a structural demand imbalance: retail and institutional traders consistently want more long exposure than short exposure, particularly in bull markets. This demand gap pushes the funding rate positive, meaning longs pay shorts. Because Ethena is always short, the protocol receives these payments. Historically, funding rates in ETH and BTC perp markets have averaged positive over multi-year periods. During periods of high market activity, funding rates can be significantly positive, generating substantial yield. During bear markets or periods of risk-off sentiment, funding can flip negative, which is the primary risk to sUSDe yield.

Source 2: Ethereum staking rewards

Where stETH is used as collateral, it continues earning Ethereum consensus and execution layer staking rewards (typically 3-6% APY) even while it serves as hedge collateral in OES custody. This staking yield is additive to funding income and provides a baseline return that exists even when funding rates are low.

Source 3: Liquid stablecoin yield

When market conditions favor a larger stablecoin allocation (low or negative funding rates), backing assets are shifted into USDC earning Coinbase's fixed rate, or into USDtb, Ethena's own stablecoin backed by BlackRock's BUIDL tokenized money market fund. These allocations earn short-term US Treasury yields, providing a floor on protocol revenue even in adverse funding environments.

sUSDe: the staking receipt

Users who want to earn yield stake their USDe to receive sUSDe. The staking contract is an ERC-4626 vault: sUSDe appreciates in value against USDe over time as protocol revenue accrues. Users do not receive yield payments directly; instead, the exchange rate between sUSDe and USDe increases with each weekly distribution cycle. To realize yield, a user simply unstakes sUSDe and receives more USDe than they originally deposited.

Approximately half of circulating USDe supply was staked to sUSDe as of October 2025. sUSDe has been deeply integrated into DeFi as a yield-bearing collateral asset,

with major protocols like Aave, Morpho, and Pendle accepting it.

The Hedging System

The delta-neutral position described in Section 3 does not maintain itself. Every mint and redemption requires a corresponding hedge adjustment. Price movements require ongoing position monitoring. Exchange outages require failover routing. This operational infrastructure is what Ethena calls the Hedging System.

The Hedging System is a set of off-chain application services that run continuously and perform the following functions:

- Ingest and normalize market data from multiple centralized exchanges, OES providers, and oracle networks (Pyth, Redstone).
- Calculate the risk exposure of the total portfolio in real time relative to defined risk limits.
- On each mint or redemption request, determine the optimal exchange for routing the hedging trade, incorporating fee structure, order book depth, existing position concentrations, and funding rate differentials.
- Calculate and publish indicative and firm prices for minting and redemption.
- Reconcile internal position records against exchange and OES provider records.
- Orchestrate the movement of backing assets between OES custody and exchange margin accounts as needed to settle P&L;
- Monitor all dependencies (exchanges, custodians, internal services) and alert the team to anomalies.

The system's guiding principle, as stated in Ethena's own documentation, is that the preservation of protocol collateral is the most important consideration. Every routing and execution decision is evaluated through a risk-first lens. The Hedging System interacts directly with both on-chain smart contracts and the Ethereum blockchain, bridging the on-chain issuance layer with the off-chain derivatives markets.

Exchange concentration is actively managed. To reduce single-point-of-failure risk,

positions are distributed across multiple derivatives venues. No single exchange holds a disproportionate share of the hedging book.

The Reserve Fund

The Reserve Fund is an additional buffer maintained by the protocol to cover funding shortfalls and to act as a buyer of last resort for USDe in open markets if the price deviates significantly from \$1.

Purpose

The primary risk to sUSDe yield, and indirectly to USDe's backing, is a sustained period of negative funding rates. If funding rates go deeply negative and stay there long enough that the staking yield on backing assets cannot offset the funding cost, the protocol would pay out more than it earns. The Reserve Fund absorbs this shortfall, ensuring that sUSDe holders do not receive negative returns and that the backing of USDe is not diluted.

Composition and security

The Reserve Fund holds liquid stablecoin assets (primarily USDC, USDT, and USDtb) denominated in the same currency as the funding payments it covers. A smaller ETH allocation covers funding on ETH-margined contracts. The fund is controlled by a 4-of-10 multi-signature wallet, with keys held by Ethena Labs contributors. As of Q4 2024, the fund held \$46.6 million. As of early 2026, it held over \$41 million in USDtb. The fund is capitalized with a portion of ongoing protocol revenue, allowing it to grow proportionally as USDe supply increases.

Open market operations

If USDe's open market price diverges far enough from \$1 in an extreme scenario where normal arbitrage is insufficient, the Reserve Fund can act as a buyer of USDe in secondary markets. Because the solvency of USDe's backing is transparently verifiable on-chain at all times, a significant deviation from peg represents an arbitrage opportunity that the Reserve Fund can exploit while simultaneously supporting the

peg.

Risk Analysis

Every yield-generating system carries risk. Ethena's design acknowledges this explicitly. The following are the primary risk categories and how the protocol addresses each.

Risk	Description	Mitigation
Funding Risk	Perpetual funding rates turn persistently negative, costing the protocol more than it earns.	Reserve Fund absorbs shortfalls. Dynamic allocation to liquid stablecoins when funding is unfavorable.
Exchange Counterparty Risk	A derivatives exchange becomes insolvent or is hacked, putting hedging positions at risk.	Off-Exchange Settlement custody means collateral is never held on an exchange. Only P&L; in transit is exposed.
Custodial Risk	OES custodian fails or is compromised.	Multiple OES providers used. No single custodian holds all assets. Institutional-grade security standards.
Backing Assets Risk	The value of backing collateral falls faster than the short hedge can offset (gap risk in an extreme crash).	Positions are monitored in real time. Margin buffers maintained. Emergency procedures can pause minting.
Stablecoin Risk	USDC or USDT depegs, affecting the value of stable reserves.	Reserve diversification across multiple stablecoins and the use of USDtb (backed by BlackRock BUIDL) as primary reserve.

Smart Contract Risk	Bugs in the on-chain contracts allow unexpected minting or draining of funds.	Multiple independent audits completed. Bug bounty program active. Contracts upgradeable only under strict governance.
Oracle Risk	Price feeds used for settlement are manipulated or stale.	Multiple oracle providers (Pyth, Redstone). Cross-validation logic. Circuit breakers in the Hedging System.

Table 2: Risk matrix

A key risk that Ethena cannot fully eliminate is what might be called systemic derivatives liquidity risk. If the notional value of Ethena's short positions grows large enough relative to the total open interest in ETH and BTC perp markets, the protocol's own hedging activity could influence funding rates, reducing yield and increasing slippage on large mints and redemptions. This is a scaling constraint that the protocol manages through multi-venue distribution and the gradual addition of new collateral types and derivatives venues.

Governance and the ENA Token

ENA is the governance token of the Ethena protocol. Its holders vote on protocol parameters including risk limits, collateral acceptance criteria, reserve fund allocation, fee switch activation, and protocol upgrades.

ENA can be staked to receive sENA, a liquid staking receipt that can be used composably across DeFi. A fee switch mechanism, proposed but not yet activated as of this writing, would divert a portion of protocol revenue to ENA stakers or toward a buy-and-burn program once certain growth milestones are met.

ENA governance does not affect USDe's peg mechanics directly. Changes to the minting contracts or hedging parameters that could affect USDe's backing are subject to timelocks, and the protocol's on-chain transparency allows any market participant to

verify the full backing of USDe in real time regardless of governance activity.

In April 2025, Germany's financial regulator BaFin ordered Ethena's German entity to cease operations, citing MiCA compliance issues. The protocol continued operating through its other corporate entities, and USDe's on-chain functions were uninterrupted. The incident highlights that regulatory risk remains a live concern for synthetic dollar protocols operating across multiple jurisdictions.

Conclusion

USDe represents the first large-scale deployment of a delta-neutral synthetic dollar that has reached meaningful adoption. By growing to the third-largest stablecoin by market capitalization in 2025, the protocol has demonstrated that the mechanism works at scale under normal market conditions.

Its key innovations are structural. The use of off-exchange settlement custody solves the exchange counterparty risk that would otherwise make the design untenable. The three-source yield model (funding rates, staking rewards, and stablecoin returns) creates resilience against any single yield source drying up. The ERC-4626 sUSDe vault design plugs cleanly into existing DeFi infrastructure. Each of these choices reflects a considered response to the specific failure modes of prior stablecoin designs.

The residual risks are real and should not be minimized. Sustained negative funding rates would test the Reserve Fund's depth. Extreme market dislocation could create temporary backing gaps. Regulatory action in key jurisdictions remains an open variable. The protocol's dependence on centralized derivatives exchanges for hedging makes it a CeDeFi hybrid rather than a fully decentralized system, and that dependence carries counterparty risk that OES custody mitigates but does not eliminate entirely.

For the DeFi ecosystem, USDe's significance extends beyond its own supply numbers. It has shown that crypto-native yield, generated without fiat bank accounts or governance token reflexivity, can be delivered reliably at scale. The design patterns it employs are likely to inform the next generation of synthetic asset protocols.

This litepaper is written for informational and portfolio purposes. It is not financial advice and does not constitute a solicitation to buy or hold any asset. All figures cited are sourced from public protocol documentation, on-chain data, and third-party research as of Q4 2025. Protocol parameters and market conditions change. Refer to the [official Ethena documentation](#) for current data.